

Abstract

This paper presents an analytical approach to linearizing a specific case of Bernoulli differential equation using geometric coordinate mapping. We demonstrate how to define a new degree of freedom within the coordinate space to derive a necessary constraint to force linearity. This method is then applied to the Bernoulli differential equation by exploiting the Chain Rule to lift the system into a higher-dimensional space. The outcome of this framework yields a solution that is robust and consistent with linear system theory.

1 Introduction

The field of Differential Equations is intrinsically messy, and finding a solution curve that satisfies any such equation can be near-impossible to achieve at times. In a non-linear system, the space is so heavily distorted that trying to unravel it directly becomes next to impossible.

When looking for a solution, it is invaluable when that system is linear. Linearity allows us to utilize linear algebra, as differential equations can often be studied through the use of linear algebra when the underlying operator is linear. However, one must ask, “what do we do when a system isn’t linear?” Or “how would we even begin to approach it?”

To tackle these questions, we must first introduce our non-linear differential equation:

$$\frac{dy}{dx} + y = y^2. \quad (1)$$

We know this equation is non-linear due to the fact that it fails to uphold additive and multiplicative invariance. This behavior violates the principle of superposition, a topic we will cover more in-depth later on in the next section. Because the y^2 term violates these fundamental properties, we cannot directly apply any standard linear frameworks. Consequently, we do not yet know whether the system can be remapped to a linear coordinate space. If we hope to achieve that, it is crucial that we understand exactly what it is we are trying to do.

The general concept is that we define a new variable, dependent on y , such that when substituted into the equation, the trajectory becomes a linear path.

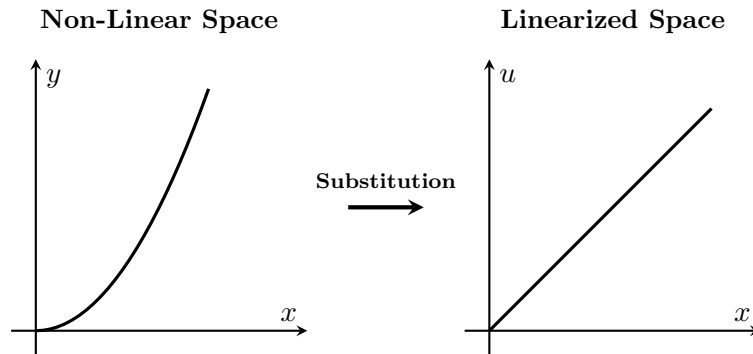


Figure 1: Visualizing the mapping of a curve from non-linear to linear via a substitution.

We can observe this simplicity in a basic algebraic case:

$$x + y^2 = 9 \quad \xrightarrow{u=y^2} \quad x + u = 9. \quad (2)$$

This substitution effectively maps our non-linear function to uphold a linear relationship. However, a major problem occurs when the equation contains multiple states of the same dependent variable that are linearly independent of one another. An example of this would be:

$$x + y^2 = y. \quad (3)$$

Here, our substitution of $u = y^2$ no longer functions to straighten the space. To circumvent this issue, we must move away from manually selecting a transformation and instead let the math force one into existence.

2 Methodology

Let's bring back our initial differential equation:

$$\frac{dy}{dx} + y = y^2. \quad (4)$$

We can assume that there must exist some function of y , denoted $u(y)$, that forces linearity along its trajectory when substituted into the system. In isolation, this new function has virtually no explicit relation to our equation. If we hope to map the system's y -axis onto $u(y)$, we must first understand how they relate to each other.

While it may not seem like there is a path at first, there is exactly one explicit relationship between y and $u(y)$ that we can exploit: their rates of change relative to x . Because the value of u depends on y , which inherently functions within the domain of x , both functions share an underlying reliance on x . Because of this shared dependency, we know that if we take the derivative of $u(y)$ with respect to x , we must also evaluate the derivative of y with respect to x . Working through these steps results in the exact logic describing u 's dynamic behavior relative to y :

$$\begin{aligned} \frac{du}{dx} &= u'(y) \frac{dy}{dx} \\ \implies \frac{dy}{dx} &= [u'(y)]^{-1} \frac{du}{dx}. \end{aligned} \quad (5)$$

Now that we understand their relation to each other, we can just substitute our new definition:

$$\begin{aligned} [u'(y)]^{-1} \frac{du}{dx} + y &= y^2 \\ \implies \frac{du}{dx} + y[u'(y)] &= y^2[u'(y)]. \end{aligned} \quad (6)$$

This mapping alone doesn't get us anywhere yet. By introducing our new function, we have added an extra degree of freedom to the system, but it remains useless until we actually impose a constraint to dictate its behavior. We can visualize this "malleability" of $u(y)$ by representing it as a curve in our (x, y, u) coordinate space:

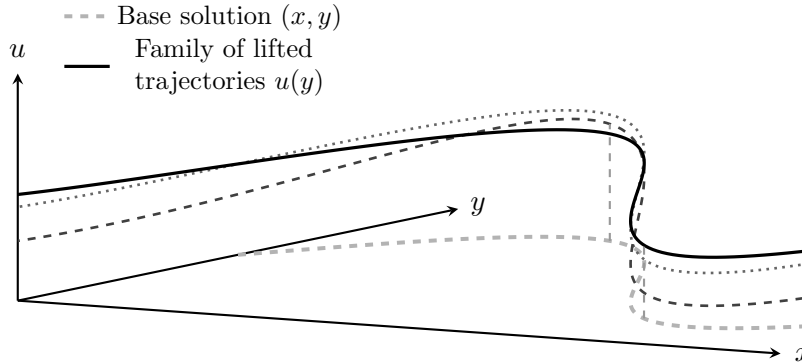


Figure 2: By introducing $u(y)$, the base (x, y) solution is lifted into 3D space. This new degree of freedom creates an infinite family of possible solutions that all share the underlying dependency on x .

Because $u(y)$ represents an infinite family of curves, we need a way to narrow it down to only one curve. The reason being: all these curves act as a solution to some specific transformation within the function space. Whether we want the system to grow exponentially, behave as a step function, or oscillate, any remapping possibility lies here. It must logically follow, then, that our specific path to a linear trajectory is embedded right here within this family. You might be asking, “how are we supposed to go from an infinite number of solutions to one single solution?” Which is a valid concern; however, we don’t need to do anything, the math does it for us. We know that we want our system to be linear, but what does that actually mean? To understand why we take our next step, we must take a look into what constraint would cause this system to become linear.

By definition, a non-linear system violates the principle of superposition (breaking multiplicative or additive invariance, as referenced in section 1):

$$T[a\mathbf{x} + b\mathbf{y}] \neq aT[\mathbf{x}] + bT[\mathbf{y}]. \quad (7)$$

If we attempt to apply a standard linear algebraic framework to our system, it quickly becomes apparent that no such linear operator can exist to satisfy our equation. We can see this simply by testing for multiplicative invariance; for any linear operator L , it must be true that $L[ay] = aL[y]$. However, if we attempt to define an operator such that $L[y] = y^2$, we find that:

$$L[ay] = (ay)^2 = a^2y^2 \quad (8)$$

while linearity strictly requires $aL[y] = ay^2$. Because $a^2y^2 \neq ay^2$ for all a , We know that no linear operator can preserve the principle of superposition when acting on a variable of non-linear degree. Therefore, for our equation to be linear, we need the left side to become a linear operator and the right side to be an expression completely independent from y .

If we take a look at the left-hand side of our system, we can define a distinct differential operator T_y :

$$\begin{aligned} T_y[y] &= \frac{dy}{dx} + y \\ \implies T_y[ay_1 + by_2] &= \frac{d}{dx}(ay_1 + by_2) + (ay_1 + by_2) \\ &= \left(a \frac{dy_1}{dx} + b \frac{dy_2}{dx} \right) + ay_1 + by_2 \\ &= \left(a \frac{dy_1}{dx} + ay_1 \right) + \left(b \frac{dy_2}{dx} + by_2 \right) \\ &= aT_y[y_1] + bT_y[y_2]. \end{aligned} \quad (9)$$

Which correctly aligns with the principle of superposition.

While we know our operator T_y behaves linearly, the right side introduces a y^2 term that forces the axis of y to scale non-linearly. Because of this, the function y within our system fails to behave as a linear entity, and therefore breaks the principle of superposition.

With this knowledge, we can now establish a constraint that attempts to resolve this issue. We know that if we want our system to be linear, the right-hand side must be independent from y . To achieve this, we can simply state:

$$y^2[u'(y)] = k. \quad (10)$$

By setting the expression dependent on y on the right side equal to a constant, we force our “malleable” function to shape itself specifically to create a valid system under this new constraint. Now,

by rearranging the equation, we end up with a definition for $u'(y)$:

$$\begin{aligned} u'(y) &= ky^{-2} \\ \implies \frac{du}{dy} &= ky^{-2}. \end{aligned} \quad (11)$$

Now look closely at what this is saying. No matter what the constant is, we will always form a linear equation. This means that, in our case for simplicity, we can define $k = 1$ giving us:

$$\frac{du}{dy} = y^{-2}. \quad (12)$$

To find out what $u(y)$ is, we can simply integrate:

$$\begin{aligned} u(y) &= \int y^{-2} dy \\ &= -y^{-1} + C. \end{aligned} \quad (13)$$

Because $u(y)$ is defined for all C , that means it must be defined for $C = 0$, which we will be using for simplicity. We can now substitute our definitions back into our transformed intermediate system:

$$\begin{aligned} \frac{du}{dx} + y \cdot y^{-2} &= y^2 \cdot y^{-2} \\ \implies \frac{du}{dx} + y^{-1} &= 1. \end{aligned} \quad (14)$$

And because of how we defined $u(y)$:

$$\begin{aligned} u(y) &= -y^{-1} \\ \implies y^{-1} &= -u(y). \end{aligned} \quad (15)$$

This allows us to say:

$$\frac{du}{dx} - u(y) = 1. \quad (16)$$

Our differential equation, in this format, is now entirely linear, maintaining scale and additive invariance.

3 Analytical Solution

This system can now be solved by treating it as a linear system of equations in the format $A\mathbf{x} = \mathbf{b}$ or more contextually precise:

$$T_u[u] = 1 \quad (17)$$

where T_u represents our newly generated linear differential operator, defined explicitly as $T_u = \frac{d}{dx} - I$. Because T_u satisfies the conditions necessary for linearity, solving this via the derivation of an integrating factor is a natural next step in computing the solution curve of $y(x)$.

By treating the equation as a linear system, we know that any solution, $u(x)$, must be in the form $u(x) = u_p + u_h$ where u_p is the particular solution and u_h is the homogeneous solution. We can then solve for the homogeneous solution by setting the right side equal to zero resulting in the following sequence:

$$\begin{aligned} \frac{du}{dx} - u &= 0 \\ \implies \frac{du}{dx} &= u \\ \implies \frac{1}{u} du &= dx. \end{aligned} \quad (18)$$

Integrating this equation yields:

$$\begin{aligned}
\ln |u| &= x + C_1 \\
\implies |u| &= e^{C_1} e^x \\
\implies u_h &= C e^x \quad \text{where } C = \pm e^{C_1}.
\end{aligned} \tag{19}$$

Now that we have isolated our homogeneous solution, u_h , we must account for our non-zero forcing function. Because we are within a linear function space, reaching this forcing function can be achieved in two ways: either by adding an independent particular solution ($u = u_h + u_p$), or by varying our constant multiplier as a function of x ($u(x) = C(x)e^x$). Both methods arrive at the same outcome, with the primary difference being that the particular solution achieves this via a translational shift, while the variable parameter “skews” or “bends” the curve through the function space until it intersects our target point. For our derivation, we will allow the parameter to vary dynamically as a function of x .

Following from this reasoning, we get:

$$\begin{aligned}
u(x) &= C(x)e^x \\
\implies \frac{d}{dx} [C(x)e^x] - C(x)e^x &= 1 \\
\implies C'(x)e^x + C(x)e^x - C(x)e^x &= 1.
\end{aligned} \tag{20}$$

Simplifying yields:

$$C'(x)e^x = 1 \implies C'(x) = e^{-x}. \tag{21}$$

Integrating this results in:

$$C(x) = -e^{-x} + C_0, \tag{22}$$

which we can then substitute into $u(x) = C(x)e^x$. Which simplifies into our general solution within our linear space:

$$u(x) = C_0 e^x - 1. \tag{23}$$

Now that we know what $u(x)$ is, we can easily find out what $y(x)$ is via a simple substitution. Recalling our original definition $u = -y^{-1}$, we equate our coordinate map to our solution curve:

$$\begin{aligned}
-y^{-1} &= C_0 e^x - 1 \\
\implies y^{-1} &= 1 - C_0 e^x \\
\implies y(x) &= \frac{1}{1 - C_0 e^x}.
\end{aligned} \tag{24}$$

The equivalence between $u(y)$ and $u(x)$ stems from the property of function composition. Because u is a function of y , and y is itself a function of x , the expression $u(y(x))$ is fundamentally a function of x , denoted here as $u(x)$.

This leaves us with our final definition for $y(x)$:

$$y(x) = \frac{1}{1 - C_0 e^x} \quad \text{where } C_0 e^x \neq 1. \tag{25}$$

4 Discussion on Limitations

This solution provides a robust trajectory for the system, but due to its rational structure, where the dependent variable sits in the denominator, there exist points within the coordinate space where the

function is undefined. As established by the condition $C_0 e^x \neq 1$, solving for the values of x where the denominator equals zero yields:

$$e^x \neq \frac{1}{C_0} \quad (26)$$

$$\begin{aligned} \Rightarrow x &\neq \ln\left(\frac{1}{C_0}\right) \\ &\neq -\ln(C_0). \end{aligned} \quad (27)$$

This condition defines the exact boundary of our solution $y(x)$, describing an asymptote that shifts dynamically based on the previously derived constant of integration C_0 .

Family of Solutions & Shifting Asymptotes

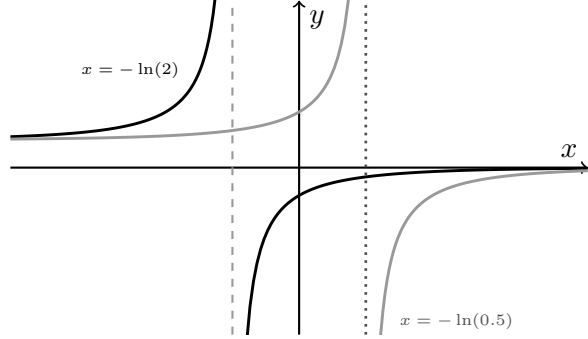


Figure 3: The integration constant C_0 acting as a coordinate translation parameter.

Another point we should touch on is the edge case:

$$u(y) = -y^{-1} \quad \text{when} \quad y = 0 \quad (28)$$

In linearizing the system, we ultimately had to sacrifice some continuity within our transformed space. This is analogous to flattening a circle; in order to do so, the surface had to be ripped apart and reshaped, so much so that it lost its topological equivalence. By performing this remapping, our coordinate transformation is forced to exclude an equilibrium point from the original system: $y = 0$.

We must also talk about the case where $C_0 < 0$. A look at the graph of $x = -\ln(C_0)$ for all C_0 reveals another condition to our solution: because of the natural log, any argument less than zero results in an undefined statement. This in turn eliminates the asymptote for any $C_0 < 0$ creating a fully continuous function defined for all x .

5 Conclusion

Solving non-linear differential equations is often done via heuristic methods rather than through actual structural derivation. By taking our time to understand and visualize the underlying “shape” of our equation, we successfully unveiled a hidden path to linearity. Through the introduction of a new dimension, $u(y)$, we were able to take advantage of the Chain Rule and force our new dimension to act as the perfect counter to our non-linear axis, y .

By imposing that our transformation coefficient be equal to some constant k , we utilize a constraint to dynamically form our “counter-dimension” such that our differential equation becomes linear. We then continued to operate on this linear equation, solving for the general solution of u using the Variation of Parameters method to yield our final explicit general solution for $y(x)$. While the concept of linear remapping is a generally robust method, it highlights a trade-off: the introduction of singularities into the transformed coordinate system. This ultimately ends up as the exclusion of an equilibrium point in the transformed system and the introduction of a conditional asymptote. Overall, this framework shows that coordinate remapping is a powerful approach to reshaping systems.